GYANISH PRAJAPATI

AEROSPACE ENGINEERING FINAL YEAR

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SUMMARY

Final-year Aerospace Engineering student at Pulchowk Engineering Campus focused on compressible aerodynamics and fluid dynamics. Hands-on experience running fluid flow simulations in ANSYS, analyzing aerodynamics in XFLR5, and building aircraft models using CATIA and Plane Maker. Proficient in using MATLAB, Python, and C to solve practical flight simulation and data-heavy engineering problems.

EDUCATION

Undergraduate 2022-present

Pulchowk Engineering Campus

- Address: Pulchowk, Lalitpur
- Subject: Aerospace Engineering
- Elective: Compressible Aerodynamics+ Rocket Propulsion+ Advanced Spacecraft Propulsion

High School 2020-2022

Khwopa Higher Secondary School

- Address: Dekocha, Bhaktapur
- Subject: +2 Science
- Additional Subjects: Math + Computer

INTERNSHIP

Nepal Academy of Science and Technology 2026 october

Research intern at Nepal Institute of Science and Technology on CUBESAT .

- Study of CUBESAT system used in BIRDS 3 .
- Design of CUBESAT mission board, Programming PIC microcontrollers and Breadboard Testing

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SKILLS

- Ansys Workbench
- Catia V5
- XFLR5
- Plane Maker
- X-Plane 12
- MATLAB and Simulink
- C-programming
- Python
- Arduino IDE
- CNC Laser Cuttter
- RDWorks V8

ACADEMIC PROJECTS

Final Year Project (In Progress)
Nano quadcopter swarm using ESP32; developing a centralized flight control architecture with a ground station, real-time wireless telemetry, and custom stabilization firmware (utilizing sensor fusion and closed-loop PID control loops).

CFD Projects (Ansys Workbench)
Performed Static Structural and Fluid Flow (Fluent) analyses using Ansys Workbench to evaluate aerodynamic performance and stress distributions.

Aircraft Manufacturing Project (CNC Laser Cutter, RDWorks V8, Catia V5)
Fabricated a Grumman G-21 model by importing and modifying open-source CAD files in Catia V5, and generating CNC laser-cutting toolpaths using RDWorks V8 for precision parts creation.

Aircraft Preliminary Design Project (X-Plane 12, Plane Maker, XFLR5)
Designed a Fixed Wing UAV that can carry medical and emergency supplies using XFLR5, Plane Maker, and X-Plane 12 to optimize aerodynamics, mass properties, and flight behavior. Validated stability, handling qualities, and flight behaviors through iterative simulation runs in X-Plane 12.

Programming Projects (MATLAB, Simulink, Python, C-programming)
Coded Numerical methods algorithms in C-Programming. Simulink modeling of MPU-6050 and PID tuning Quadcopter model for various inputs. MATLAB algorithms for CFD 1D convection diffusion schemes. Learning Python for developing the ground station software to monitor real-time wireless telemetry, log flight data, and visualize drone swarm trajectories.

REFERENCES

Asst. Prof. Biman Rimal, Department of Mechanical and Aerospace Engineering, Pulchowk Campus
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